

 Category	Component	Use	#	Qty	Cost Each	Total Cost	Specs
Compute	ESP32	Webserver interface and motor control		1	7	\$7.00	ELEGOO ESP-32 Development Board USB-C, 2.4GHz Dual Mode WiFi+Bluetooth Dual Core Microcontroller Support AP/STA/AP+STA, CP2102 Chip
Drive	RS-555 Gear Motors + Gearboxes	Main drive motors and gear reduction units		2	16	\$32.00	24 V DC; ~35 000 RPM motor; 1:100 gear ratio; ~200 W @ 24 V; steel middle gear; composite housing
Drive	BTS7960 Motor Drivers	Dual H-bridge drivers for DC motors		2	5.5	\$11.00	5.5–27 V supply; up to 43 A continuous
Drive	10" Pneumatic Tires w/ Hubs	Drive wheels		2	9	\$18.00	10" pneumatic utility tires w/ steel hub
Drive	6" Solid Rubber Caster Wheel	Rear caster		1	8	\$8.00	6" solid rubber tire
Frame	Plywood	Frame		0.5	40	\$20.00	¾" plywood panel, (36) 2" wood screws, (24) 1" wood screws
Frame	½" All-Thread	Axle shaft connecting wheels, gearboxes & frame		2	8	\$16.00	½"-13 all thread steel rod (16" length); threads partially ground to ≈12 mm for fit with 12 mm hole in gear motors
Power	12 V 7 Ah SLA Battery	Power		1	21	\$21.00	Lead-acid sealed battery with F1 terminals; maintenance-free; 7 Ah capacity
Power	DC-DC Buck Converter	Power Source for ESP32		1	2.75	\$2.75	Working Voltage: DC 9V - 36V; The Output Voltage: 5-5.3V
Power	XT60 Connectors	Power connectors for batteries and motor drivers		4	0.6	\$3.00	XT60H male/female pairs; rated 60 A; nylon shell
Power	LED Rocker Switches (12 V 20 A)	Main on/off switch		1	1	\$1.00	SPST ON-OFF; 12 V 20 A rated; blue LED indicator; 3-pin surface mount
Sensors	GY-521 MPU-6050	Get Yaw (Heading)		1	4	\$4.00	6-DOF IMU
Total							\$143.75